The syntax of extraposition: Lessons from the fourth reading

(This is currently an addendum to "Reconstruction, bare quantification, and the fourth reading", and is being developed into an independent paper)

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Reconstruction is argued to be blocked for RC adjuncts extraposed over a right-adjoined adverb. Hulsey & Sauerland (2006) illustrate this by attempting to extrapose RCs that comply with independent syntactic constraints *only* if their head reconstructs into a RC-internal position. The resulting ungrammaticality or infelicity in (1) suggests that reconstruction, required for the satisfaction of Condition A in (1a) and to achieve an idiomatic interpretation of (1b), is blocked by extraposition.

(1) a. * I saw the portrait of himself$_i$ yesterday that John$_i$ likes.[1]

b. # Mary could not ignore the tabs yesterday that John was keeping on her.

As predicted by an account that derives the fourth reading as a reconstruction effect, the reading relative to modals (henceforth, R4-M) is indeed blocked when the modal-embedding RC is extraposed. In the exclusively R4-M-verifying scenario (2a), the example without extraposition seems true (2b), whereas its counterpart with it in (2c) is judged false.

(2) a. **Context:** Alex is again awaiting the results of the judicial elections, whose candidates are again all non-judges, to see who will preside over her court case. She is a well-known politician and is encouraged by her lawyer to use her political platform to praise all judicial candidates, so that whoever wins will judge her favorably. She says:

b. I praised every judge that we could face in court. **true** in (2a)

c. I praised every judge yesterday that we could face in court. **false** in (2a)

(adapted from Benbaji 2021: ex. 27)

This falls out from the reductive account, of course, if extraposition blocks reconstruction, which is a precondition for R4-M.

1 This is Hulsey & Sauerland's example. Some speakers may judge the sentence to be better than predicted if extraposition indeed blocks reconstruction, given that animate anaphors are sometimes exempt from binding condition A (cf. Charnavel & Sportiche 2016; Charnavel 2019). To avoid this confound, we may consider the contrast in (i), where an inanimate anaphor is used.

(i) a. I noticed the citation of itself$_i$ that the paper$_i$ mentions in a footnote.

b. * I noticed the citation of itself$_i$ yesterday that the paper$_i$ mentions in a footnote.

I thank Isabelle Charnavel and Dominique Sportiche (p.c.) for this point.

We can, in principle, end the discussion of extraposition at that, with the correlation between the absence of reconstruction and the unavailability of R4-M once again vindicated. But, once we examine more closely the question of *why* extraposition should block reconstruction in the first place, we can formulate a final prediction for our analysis of R4-M, that will also have non-trivial implications for the theory of extraposition. Addressing that question requires a bit more background on the syntax of relativization, to which we turn next.

1 Why does extraposition block reconstruction?

Since Lees (1961), it is often assumed that in addition to their raising parse, RCs have a non-raising one, in which the NP-head in its surface position does not form a movement chain with any RC-internal element (cf. Carlson 1977). Abstracting away from the details, the derivation of such RCs, termed matching RCs (henceforth MRCs), differs from the raising derivation, repeated in (3) from work by Bassi & Rasin (2018), only in that it is not two NPs related by *movement* that occupy the RC-external position of the head and its hypothesized RC-internal one, but rather two independent NPs—an external *and* and internal head—with the latter phonologically deleted under *semantic identity* with the former (Sauerland 1998, 2004). Thus, neither of these NPs can be ignored at LF, which is what reconstruction requires in our raising parse. All other features of the raising construal—namely, trace conversion of the internal head and abstraction over the variable it introduces—remain in the matching structure, schematized in (4).[2]

(3) *Raising schema*

a. *Merge copy of NP ("raising")*:
$[_{RC} \ldots [_{DP}[D][_{NP}\ N]]] \rightarrow [[_{NP}\ \underset{\wedge}{N}][[\lambda_i][_{RC} \ldots [_{DP}[D][_{NP}\ \underset{\vee}{N}]]]]]$

b. *Trace conversion:*

i. $[[_{NP}\ N][[\lambda_i][_{RC} \ldots [_{DP}[D][_{NP}\ N]]]]$
$\rightarrow [[_{NP}\ N][[\lambda_i][_{RC} \ldots [_{DP}[D][_{NP}\ [N][\lambda y.y = x_i]]]]]]$

ii. $\rightarrow [[_{NP}\ N][[\lambda_i][_{RC} \ldots [_{DP}[\text{THE}][_{NP}\ [N][\lambda y.y = x_i]]]]]]$

c. *Higher copy LF-deletion ("reconstruction"):*
$[[_{NP}\ N][[\lambda_i][_{RC} \ldots [_{DP}[\text{THE}][_{NP}\ [N][\lambda y.y = x_i]]]]]]$
$\rightarrow [[\lambda_i][_{RC} \ldots [_{DP}[\text{THE}][_{NP}\ [N][\lambda y.y = x_i]]]]]$

(4) *Matching schema:* $[[_{NP}\ N][[\lambda_i][_{RC} \ldots [_{DP}[\text{THE}][_{NP}\ [N][\lambda y.y = x_i]]]]]]$
(where the higher and lower nouns are taken to have identical meanings)

2 We are greatly simplifying the matching structure in (4), by ignoring movement of the internal head hypothesized to take place RC-internally. This is of no consequence in what follows.

The main data in support of a matching *vs* raising ambiguity involves grammatical RCs whose heads contain a R(eferential)-expression that would be bound, in violation of Condition C, had it originated RC-internally (5a). The acceptability of (5a) contrasts with the alleged unacceptability of comparable *wh*-chains, where moving a R-expression from a position in which it is bound to one in which it is not, does not seem to obviate Condition C (5b).

(5) a. The picture of John$_i$ that you thought he$_i$ liked is on the mantle.

b. ?* Which picture of John$_i$ did you think he$_i$ liked?

If movement does not obviate Condition C, the RC in (5a) must have a parse in which the R-expression does not originate below its potential binder. According to Sauerland (2004), the matching parse plays this role. MRCs contain two heads, internal and external, that are *not* related by movement. And while the elements in a movement chain must be "strictly identical", the two heads in a MRC need only be "identical in meaning" (2004: 110). This flexibility allows the internal head in (5a) to contain a pronoun,[3] rather than a R-expression, and is thus not subject to Condition C; e.g., (6).

(6) The λx [picture of John$_i$] that you thought he$_i$ liked [~~the$_x$ picture of him$_i$~~].

Hulsey & Sauerland further adopt Fox & Nissenbaum's (2000) view that extraposition of a RC past a right-adjoined adverb in a structure like (7a) consists of covert-movement (i.e., QR) of a DP past the adverb (7b), followed by trace conversion of the lower copy of the DP (7c), and finally a counter-cyclic operation that lets us "late-merge" that extraposed RC directly into the higher one (7d).

(7) a. *Merger of adverb:* $[\ldots[_{\text{DP}}\ \text{D}\ [_{\text{NP}}\ \text{N}]]\ldots\text{adverb}\ldots]$

b. *"Rightward" QR of DP past adverb:*
$[[[[\ldots[_{\text{DP}}\ \text{D}\ [_{\text{NP}}\ \text{N}]]\ldots\text{adverb}\ldots]][\lambda_i]][_{\text{DP}}\ \text{D}\ [_{\text{NP}}\ \text{N}]]]$

c. *Trace conversion of lower copy:*
$[[[[\ldots[_{\text{DP}}\ [\text{THE}_i][_{\text{NP}}\ \text{N}]]\ldots\text{adverb}\ldots]][\lambda_i]][_{\text{DP}}\ \text{D}\ [_{\text{NP}}\ \text{N}]]]$

d. *"Late-merger" of RC into the DP heading the movement chain:*
$[[[[\ldots[_{\text{DP}}\ [\text{THE}_i][_{\text{NP}}\ \text{N}]]\ldots\text{adverb}\ldots]][\lambda_i]][_{\text{DP}}\ \text{D}\ [[_{\text{NP}}\ \text{N}][\mathbf{RC}]]]]$

If extraposition indeed consists of these four steps, it directly follows that the extraposed RC cannot have a raising construal. Raising requires that the head-NP be merged RC-internally and raise to its surface position. But by the time the RC is merged in (7d), that NP has already been merged and moved past an adverb, preventing it from originating in the RC-internal gap position.

3 This is known as "vehicle-change" since Fiengo & May 1994.

This approach has been challenged from two conflicting directions: On the one hand, it has been argued that the contrast taken to motivate the existence of MRCs in (5) "is not as robust or theoretically significant as it has been taken to be" (Henderson 2007: 204). In particular, the ungrammaticality judgment in (5b) has been questioned, suggesting, perhaps, that movement may obviate Condition C violations,[4] and therefore that matching is superfluous in accounting for the grammaticality of (5a).[5] Henderson thus argues that all RCs involve raising, and that we can maintain the account of extraposition in (7) by adopting a particular architecture of movement that lets us ignore the higher head-NP in a RC movement chain even when the RC had been late-merged with it.

Briefly, the proposal rests on Nunes's (2004) theory of movement, according to which elements in the syntactic numeration may be copied and merged in separate derivational workspaces, and thus not *all* copies of the same element must be in a c-command relation. Rather, a post-derivational operation of *chain-formation* identifies those copies of the same element that are in a c-command relation and marks them as a chain. A well-formedness requirement then ensures that for every chain at least one copy is interpreted (and only one is pronounced; though see Henderson 2007: fn. 18 for complications). Given the representational nature of chain-formation, QR of a DP past an adverb as in (7b), does not prevent the DP's NP restrictor from forming a chain with its copy inside the late-merged raising RC.

On the other hand, there has been a push in the literature to eliminate the raising parse altogether, and account for connectivity and anti-connectivity effects between the external head and the RC-internal environment in MRCs by positing a deletion operation that allows either the external or the internal head to remain uninterpreted at LF (cf. Munn 1994; Citko 2001; Salzmann 2019, a.o.). On this view, when the head-NP contains an anaphor or an idiom part, we are allowed to ignore its RC-external copy, but when it contains an R-expression bound RC-internally, we are allowed to ignore its RC-internal one.[6]

These views, while stark opposites, nevertheless converge on a similar structure for extraposed RCs, schematized in Figure 1, where copies are boldfaced for readability, and numbered to facilitate reference below. QR of DP_γ past an adverb forms the movement chain $\{DP_\gamma\text{-}1, DP_\gamma\text{-}2\}$, whereas NP_β-1 and NP_β-2 form their own

4 This claim, originating in Safir 1999, received experimental support in Bruening & Al Khalaf 2019.

5 Henderson (2007), following Safir (1999), suggests that "vehicle-change" is a general property of *wh*-chains, allowing the lower and higher copies to differ in whether they contain a pronoun or a R-expression, though no comprehensive theory of vehicle change is provided.

6 Most proponents of matching seem to disagree with Henderson (2007) on the robustness and significance of the Condition C data in (5), but this issue is beyond our immediate interests below. Note, also, that the matching-only approach is not in conflict with our account of R4-M which is presented above as contingent on the raising construal of RCs, given that proponents of matching must posit an operation that mimics the effects of reconstruction for independent reasons.

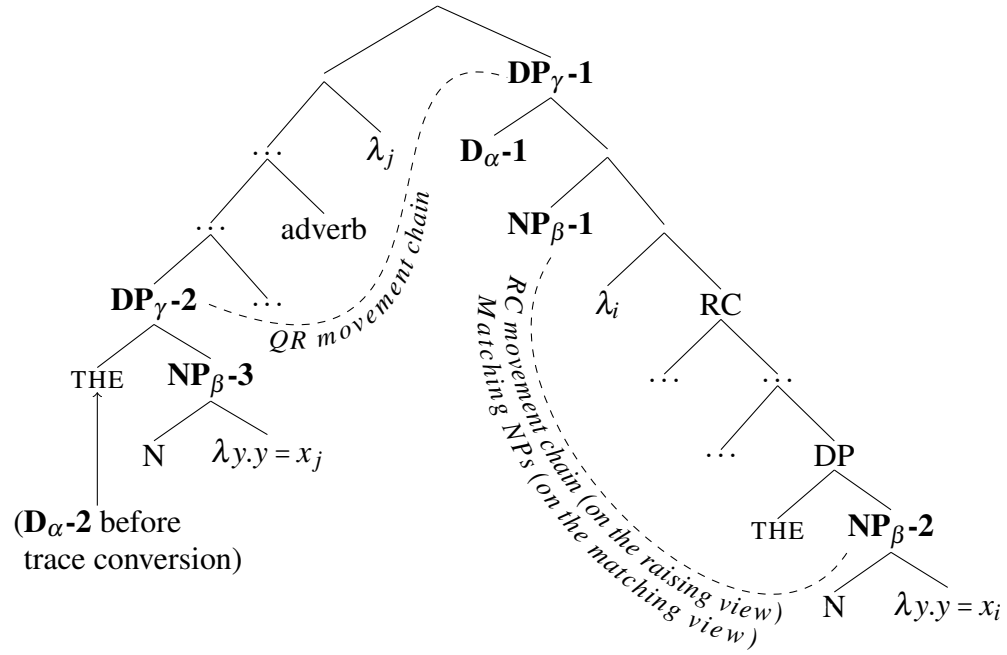


Figure 1 Structure for extraposition with a raising RC

movement chain on the *raising-only* approach, but are not related by movement on the *matching-only* approach. Either way, however, we are allowed to ignore the higher of the two, NP$_\beta$-1, and interpret NP$_\beta$-2 at LF, which is what is required to achieve the effects of reconstruction. If that is the case, of course, we are in need of an alternative explanation for the apparent incompatibility between extraposition and reconstruction, as Hulsey & Sauerland's claim that a late-merged RC cannot in principle show reconstruction effects no longer holds.

This apparent incompatibility, recall, is demonstrated by the ban on extraposing RCs whose head-NP contains an anaphor bound RC-internally (8a), or the absence of an idiomatic interpretation when the head-NP forms an idiom with RC-internal material (8b); i.e., instances of the schema in Figure 1 in which NP$_\beta$ contains an anaphor or idiom part, whose binder or idiomatic environment is RC-internal.

(8) a. * $[\ldots[_{\text{DP}}\ \text{D}\ [_{\text{NP}}\ \ldots\ \text{anaphor}_i\ \ldots]_j]\ldots\ \text{adverb}][_{\text{RC}}\ldots\ \text{R-expression}_i\ \ldots\ \text{t}_j]$

b. # $[\ldots[_{\text{DP}}\ \text{D}\ [_{\text{NP}}\ \text{idiom part}]_j]\ldots\ \text{adverb}][_{\text{RC}}\ldots\ \text{idiomatic environment}\ \text{t}_j]$

Condition A and the constraint maintaining idiom integrity can be thought of as "licensing conditions" on such NPs. And if NP$_\beta$ in Figure 1 is subject to one of these conditions, the *only* copy of it that is licensed is NP$_\beta$-2, as both NP$_\beta$-1 and NP$_\beta$-3 do not comply with these constraint: If NP$_\beta$ contains an anaphor, for instance, that anaphor is bound only in the position of NP$_\beta$-2. Similarly, if NP$_\beta$

contains a part of a VP idiom, NP_β-1 and NP_β-3 simply will not have an idiomatic meaning as they do not reside in the complement position of an idiomatic verb. Thus, only NP_β-2 may be interpreted.

But, Henderson (2007) suggests, therein lies the rub: When only NP_β-2 is interpreted, no NP in the QR chain, and by extension no DP forming that chain, is interpreted. Thus, as long as we posit a constraint demanding that at least one element of every movement chain be interpreted, we derive the ungrammaticality in (8) without conceding that the RCs in that schema cannot show reconstruction effects. The source of ungrammaticality, i.e., the "offending" NP, is not part of these RCs. Rather, it is NP_β-3, at the tail of the QR chain, which violates the licensing constraint that reconstruction effects within the RC are supposed to circumvent.

A similar story can explain the apparent incompatibility of R4-M and extraposition demonstrated with (9a), which seems *false* in the exclusively-R4-M-verifying scenario (2a). Here, the need to be interpreted RC-internally, while not a *licensing* constraint on the head-NP, is required to render the sentence true in its respective context. So while in (9a), too, the *judge*-NP heading the RC can reconstruct to receive opaque evaluation, its trace-converted copy within the QR chain, boldfaced in (9b), must be interpreted, and given that it is not in the scope of *any* intensional operator, it must be interpreted *transparently*, "undoing" the effects of reconstruction in the RC. The sentence with extraposition, paraphrased in (9c), is thus committed to the existence of *actual* judges, explaining the absence of R4-M.

(9) a. I praised every judge today that we could face in court.

b. [λ_y I praised **the**$_y$ **judge** today][every ~~judge~~ [λ_x we could face the$_x$ judge]]
QR chain; RC external and internal NPs

c. Every individual x for whom there is a possible world in which we face x as the judge in court, is such that I praised the actual judge x today.

Henderson's (2007) explanation for the infelicity in (8) is not the consensus view.[7] However, it makes a prediction regarding the interaction between extraposition and reconstruction that is applicable to R4-M, and is more nuanced than Hulsey & Sauerland's (2006) absolute ban on reconstruction in extraposed RCs. Reconstruction is not predicted to be blocked for *all* extraposed RCs, but only for those whose head-NP is not independently licensed within the QR chain in the matrix clause. If it is licensed, extraposition should be possible, as corroborated by contrasts like (10-11), where the head-NP is licensed within the QR chain.

(10) a. * Mary saw the portrait of himself$_i$ yesterday that John$_i$ likes.

b. John$_i$ saw the portrait of himself$_i$ yesterday that he$_i$ likes.

7 See Salzmann 2006, 2019 for a discussion.

(11) a. # Mary could not ignore the beans yesterday that John spilled.

b. Mary spilled the beans yesterday that she thought John already spilled.

(based on Henderson 2007: ex. 28)

Next, we spell out what this view of the interaction of extraposition and reconstruction predicts for R4-M, on the assumption that R4-M is a reconstruction effect.

2 A more nuanced prediction regarding R4-M and extraposition

R4-M involves cases in which a determiner seems to scope above a modal, but a NP adjacent to it is evaluated in the modal's scope. When the modal is RC-internal, the determiner under which the RC is embedded was never merged in the scope of the modal to begin with, and the adjacent NP may reconstruct, deriving the desired "split-scope" effect. In extraposition cases, the determiner and adjacent NP first form a DP that QRs, and only then merge with the intensional RC. If the absence of R4-M in such cases is really due to the copy of the NP within the QR chain being interpreted outside the scope of a modal, we predict that placing a similar modal above it in the matrix clause will render extraposition compatible with R4-M. This is because, trace-conversion within the QR chain will effectively remove the quantificational force of the lower determiner from the scope of the matrix-clause modal (cf. D_{α}-2 in Figure 1), but its NP complement will nevertheless be evaluated opaquely, as required for R4-M.

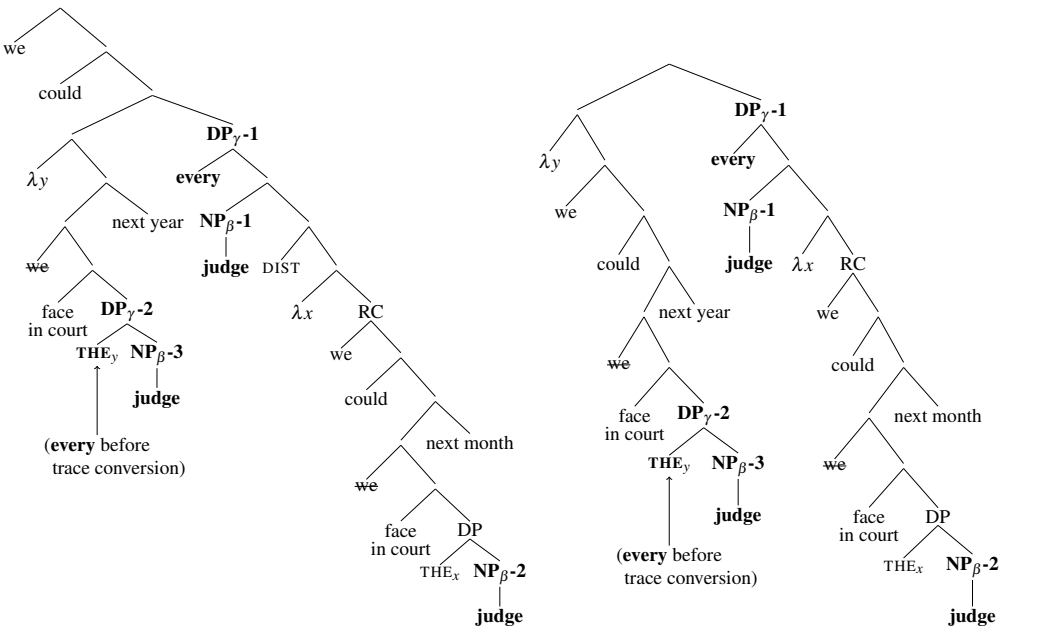
We illustrate that this prediction is borne out with the sentence in (12). Which contains a RC extraposed over an adverb, with an occurrence of an existential modal in both the RC and the matrix clause.

(12) We could face in court every judge next *year* that we could next *month*.

On Fox & Nissenbaum's (2000) account, extraposition requires QR of a matrix-clause DP prior to the late-merger of the extraposed RC. Thus, (12) can have at least two structures, both schematized in Figure 2 below; one in which the DP "*every judge*" QRs to a position below the matrix "*could*" (Figure 2-a), and another in which it QRs to a position above it (Figure 2-b).[8]

Regardless of which copy of the head-NP is interpreted in the late-merged RC, Figure 2-a gives rise to a reading on which it is possible for Alex to face in court

8 The two copies of the QRed DP are boldfaced in Figure 2, as are the internal and external head-NPs of the RC, which can be thought of as matching NPs or as copies in a movement chain, either works for our purposes as discussed above. As is standard, the subject in both structures "*we*" is assumed to originate below the modal, where it is interpreted, but is pronounced in a higher position. Both positions are represented in the figure, with the unpronounced copy crossed-out. Note, futher, that we freely insert distributivity operators throughout the structure.



(a) LF that derives a *de dicto* reading (b) LF that derives *R4-M* or *de re*

Figure 2 Possible structures for (12)

every individual who could be a judge; i.e., a *de dicto* reading of the raised DP relative to the matrix modal, paraphrased in (13a). Crucially, however, if extraposed RCs can show reconstruction effects, Figure 2-b is predicted to give rise to R4-M, paraphrased in (13b). This is because both NP$_\beta$-2 and NP$_\beta$-3 are in the scope of modals of a similar force and flavor in that figure, and thus even when both are interpreted, the resulting proposition is not necessarily committed to the existence of actual judges. And since trace-conversion of the matrix DP effectively removes the quantificational force of the determiner of DP$_\gamma$-2, the only quantificational determiner in the structure—namely, the one in DP$_\gamma$-1—scopes above both modals. Thus, the interpreted NPs can be evaluated in the domain of modals which the quantificational determiner outscopes. Note that if Hulsey & Sauerland (2006) are correct in arguing that extraposed RCs cannot show reconstruction effects, Figure 2-b is predicted to give rise only to a *de re* reading, paraphrased in (13c), which is committed to the existence of actual judges.

(13) a. The following is possible: we face in court next year all individuals who could be the judge we face in court next month. (*de dicto*)

b. Every individual who could the judge we face in court next month, could also each be the judge we face in court next year. (*R4-M*)

c. Every actual judge we could face in court next month, could also be the judge we face in court next year. (*de re*)

Now consider (12) in scenario (14). Since—regardless of when exactly her lawsuit will be filed—Alex will eventually face only *one* judge in court, the *de dicto* reading (13a) is false. And since there are no relevant *actual* judges in the scenario, so is the *de re* reading (13c). The scenario verifies, exclusively, the R4-M of (12), and the fact that the sentence does, indeed, seem felicitous and true in this scenario is a striking corroboration, we think, of Henderson's (2007) position *vis-a-vis* the interaction of extraposition and reconstruction, and its implications for R4-M, on the assumption that the latter phenomenon is a reconstruction effect.

(14) **Context:** Ten vacancies in the local court will be filled in the upcoming judicial elections, whose candidates are not and never were judges. The ten eventual winners will stay on the court for at least two years. Alex intends to file a lawsuit, which will be decided by just one of the ten winners, selected at random. She is certain that the candidates in the current election cycle are likelier to accept her lawsuit than anyone who will occupy their seats in the future. She therefore pressures her lawyer to file the lawsuit as soon as next month, knowing this process takes time, and fearing she will miss her opportunity to face a friendly judge. Her lawyer wants to postpone the process until next year, and would like to calm Alex's fears from this delay. He knows that Alex is not aware of the expected lengths of the winners' tenures, and uses (12) to convey that a yearlong delay will not impact the identity of the presiding judge.

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